Robotic Autonomous Table Swarm

The Chevrotains
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Background

Setting and clearing a family table can be unsanitary and repetitive. The Robotic Autonomous Table Swarm (RATS) uses swarm robotics strategies to automate this task.

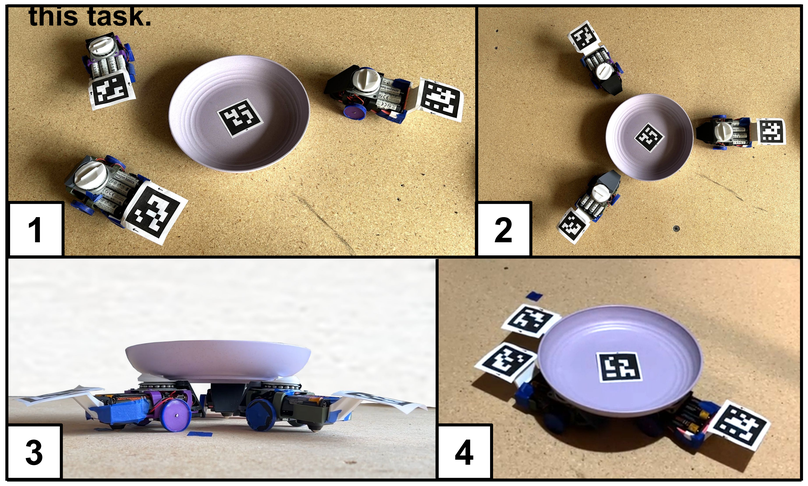


Figure 1: The RATS use an overhead camera to (1) starting in any position, (2) navigating to equidistant points around the plate, (3) lifting the plate, and (4) carrying the plate to a target position

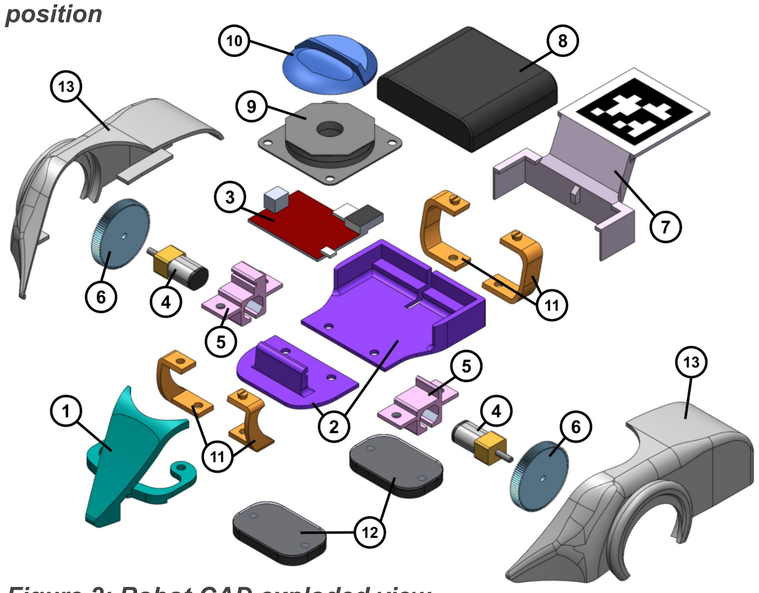


Figure 2: Robot CAD exploded view

Part Labels:

1. Ramp
2. Base Plate
3. Main Circuit Board
4. PMDC Motors
5. Motor Mounts
6. Wheels
7. ArUco Tag Mount
8. Battery Pack (4-AA)
9. Turntable
10. Turntable Top
11. Turntable Mounts
12. Roller Ball Caster
13. Enclosure

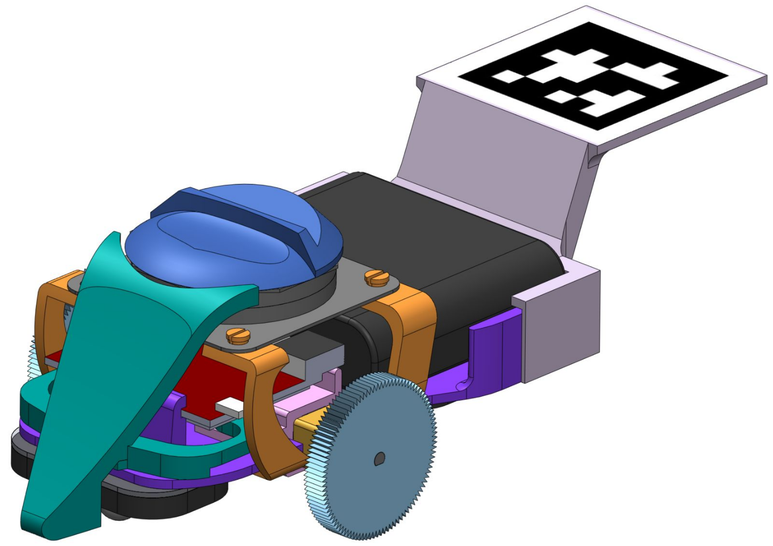


Figure 3: Robot CAD assembly (enclosure hidden)

Design of Robot

Our biggest design challenge was finding a way to fit all three robot ramps underneath the plate during lifting. To solve this, we compacted the design, minimizing turntable to ramp distance and lowering overall robot height.

Testing & Modeling	Target	Model	Actual	F.O.S.
Robot Weight (g)	367	279	279	N/A
Plate Weight (g)	300	228	116	1.96
Motor Strength (kg cm)	1.06	0.268	2	7.46

Grid Making and Multi-Robot Pathfinding

A key task was finding a collision-free path from start to finish. Our A* pathfinding algorithm treats other robots as dynamic obstacles, checking new potential routes against the other robot paths in space and time. It also minimizes a cost function for each of the paths with different weights on translational and rotational motion.

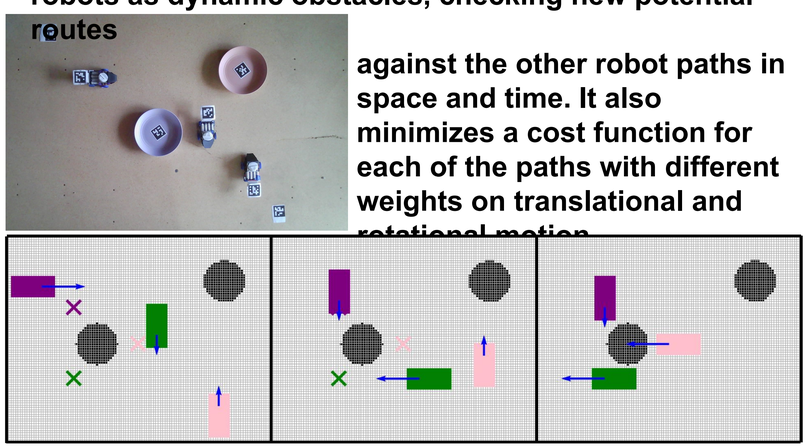


Figure 4: Converting a camera image into valid robot paths

Live Object Avoidance

Pathfinding avoids obstacles in theory, but real-world dynamics can cause collisions with plates or other robots. Before each command is sent, the laptop runs a collision prediction model — if a risk is detected, evasive routines are sent to each robot.

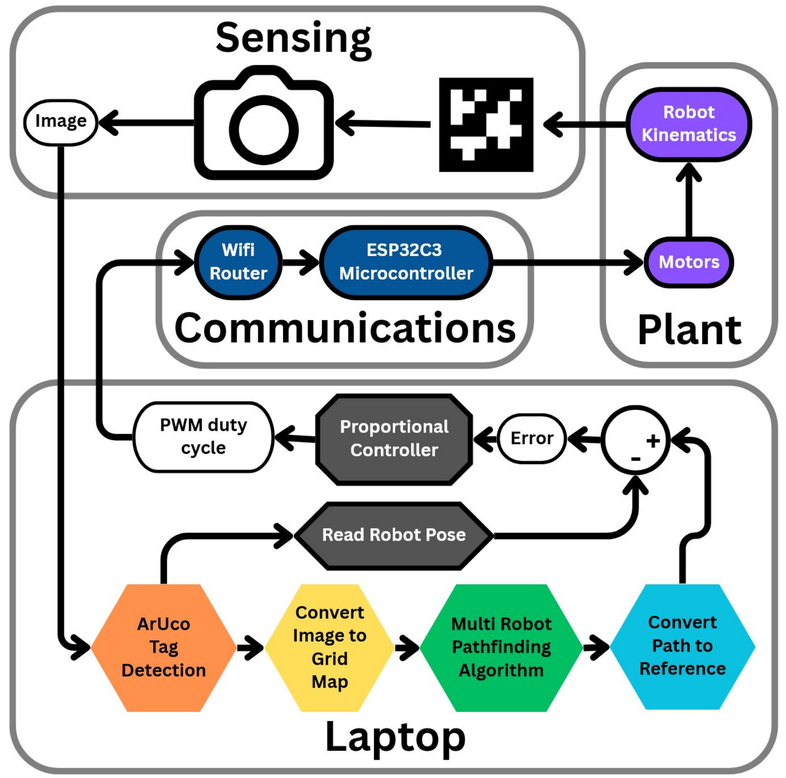


Figure 5: The system flowchart, shows how information is passed and processed during operation

Future Work

To further expand upon this project, we could replace ArUco tag detection with machine vision, eliminating the tag mounts and plate tags. The drivetrain could be upgraded to an omni wheel drive with stronger motors, to better move the plate freely and be able to lift heavier, ceramic plates. We could also shift our software onto ROS2 to make the system more robust.

Acknowledgments

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References

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Spencer, Matthew J. “Pathfinding - Understanding A* (A Star).” YouTube.